

Настройка VPN

Лабораторная работа №16

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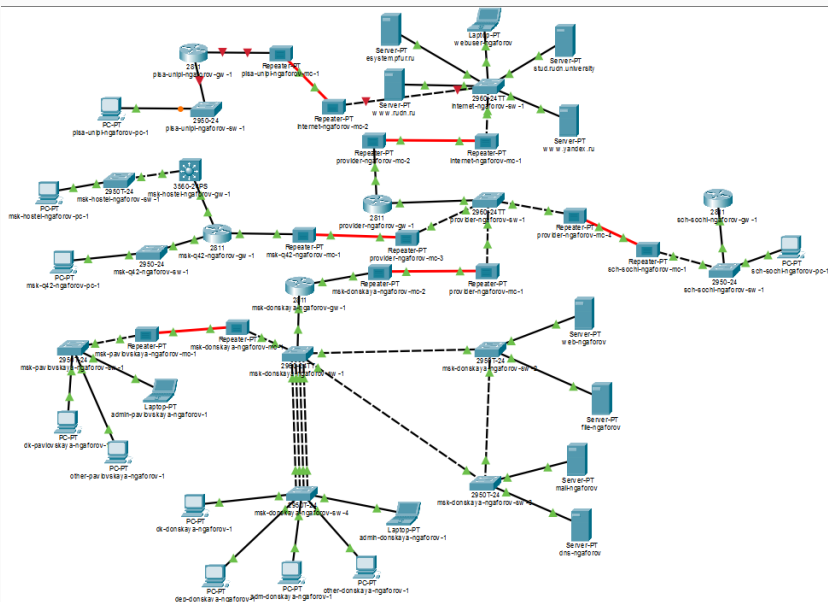
Российский университет дружбы народов, Москва, Россия

Цель работы

Получение навыков настройки VPN-туннеля
через незащищённое Интернет-соединение.

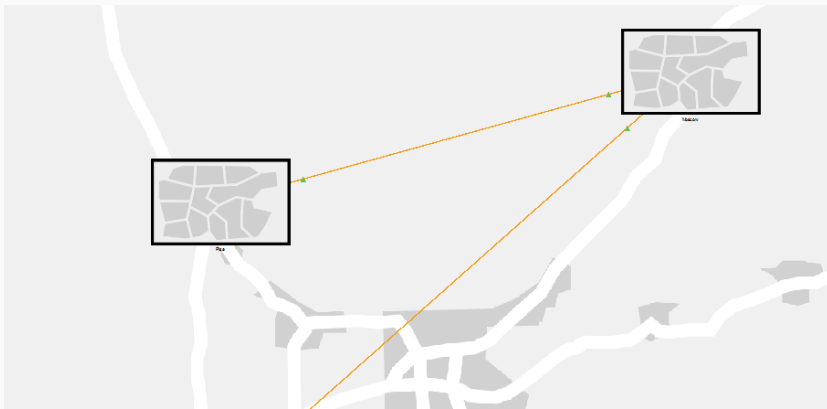
Ход выполнения

Общая топология проекта



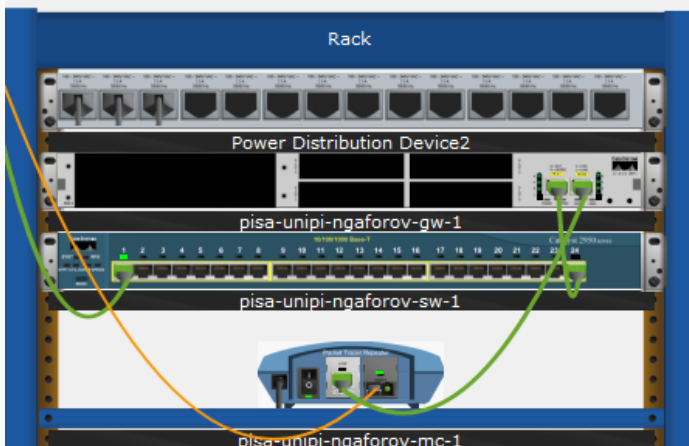
Физическая область проекта

- Создан город **Pisa**
- Добавлено здание **University of Pisa**
- Оборудование перенесено в соответствующую физическую область
- На карте отображены удалённые площадки и линии связи



Размещение оборудования в стойке

- Проверено расположение устройств в здании Университета г. Пиза
- В стойке размещены маршрутизатор, коммутатор и устройство связи
- Соединения соответствуют логической топологии



Первоначальная настройка маршрутизатора

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#host pisa-inipi-ngaforov-gw-1
pisa-inipi-ngaforov-gw-1(config)#line vty 0 4
pisa-inipi-ngaforov-gw-1(config-line)#pass cisco
pisa-inipi-ngaforov-gw-1(config-line)#login
pisa-inipi-ngaforov-gw-1(config-line)#exit
pisa-inipi-ngaforov-gw-1(config)#line console 0
pisa-inipi-ngaforov-gw-1(config-line)#pass cisco
pisa-inipi-ngaforov-gw-1(config-line)#login
pisa-inipi-ngaforov-gw-1(config-line)#exit
pisa-inipi-ngaforov-gw-1(config)#enab secret cisco
pisa-inipi-ngaforov-gw-1(config)#serv pass
pisa-inipi-ngaforov-gw-1(config)#user admin priv 1 secret cisco
pisa-inipi-ngaforov-gw-1(config)#ip domain-name unipi.edu
pisa-inipi-ngaforov-gw-1(config)#crypto key gen rsa
The name for the keys will be: pisa-inipi-ngaforov-gw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

pisa-inipi-ngaforov-gw-1(config)#line vty 0 4
*Mar 1 0:6:9.769: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:6:9.769: %SSH-5-ENABLED: SSH 1.5 has been enabled
pisa-inipi-ngaforov-gw-1(config-line)#transport in ssh
pisa-inipi-ngaforov-gw-1(config-line)#exit
pisa-inipi-ngaforov-gw-1(config)#
```


Первоначальная настройка коммутатора

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#host pisa-unipi-ngaforov-sw-1
pisa-unipi-ngaforov-sw-1(config)#line vty 0 4
pisa-unipi-ngaforov-sw-1(config-line)#pass cisco
pisa-unipi-ngaforov-sw-1(config-line)#login
pisa-unipi-ngaforov-sw-1(config-line)#exit
pisa-unipi-ngaforov-sw-1(config)#line console 0
pisa-unipi-ngaforov-sw-1(config-line)#pass cisco
pisa-unipi-ngaforov-sw-1(config-line)#login
pisa-unipi-ngaforov-sw-1(config-line)#exit
pisa-unipi-ngaforov-sw-1(config)#enab secret cisco
pisa-unipi-ngaforov-sw-1(config)#serv pass
pisa-unipi-ngaforov-sw-1(config)#user admin priv 1 secret cisco
pisa-unipi-ngaforov-sw-1(config)#ip domain-name unipi.edu
pisa-unipi-ngaforov-sw-1(config)#crypto key gen rsa
The name for the keys will be: pisa-unipi-ngaforov-sw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

pisa-unipi-ngaforov-sw-1(config)#line vty 0 4
*Mar 1 0:7:46.54: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:7:46.54: %SSH-5-ENABLED: SSH 1.5 has been enabled
pisa-unipi-ngaforov-sw-1(config-line)#trans input ssh
pisa-unipi-ngaforov-sw-1(config-line)#exit
pisa-unipi-ngaforov-sw-1(config)#
```

Настройка интерфейсов маршрутизатора

```
pisa-inipi-ngaforov-gw-1(config)#
pisa-inipi-ngaforov-gw-1(config)#int f0/0
pisa-inipi-ngaforov-gw-1(config-if)#no sh

pisa-inipi-ngaforov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

pisa-inipi-ngaforov-gw-1(config-if)#exit
pisa-inipi-ngaforov-gw-1(config)#int f0/0.401
pisa-inipi-ngaforov-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up

pisa-inipi-ngaforov-gw-1(config-subif)#enc dot1q 401
pisa-inipi-ngaforov-gw-1(config-subif)#ip addr 10.131.0.1 255.255.255.0
pisa-inipi-ngaforov-gw-1(config-subif)#descr unipi-main
pisa-inipi-ngaforov-gw-1(config-subif)#exit
pisa-inipi-ngaforov-gw-1(config)#int f0/1
pisa-inipi-ngaforov-gw-1(config-if)#no sh

pisa-inipi-ngaforov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

pisa-inipi-ngaforov-gw-1(config-if)#ip addr 192.0.2.20 255.255.255.0
pisa-inipi-ngaforov-gw-1(config-if)#descr internet
pisa-inipi-ngaforov-gw-1(config-if)#exit
pisa-inipi-ngaforov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 192.0.2.1
pisa-inipi-ngaforov-gw-1(config)#
```

Настройка VLAN на коммутаторе

```
pisa-unipi-ngaforov-sw-1(config)#
pisa-unipi-ngaforov-sw-1(config)#int f0/24
pisa-unipi-ngaforov-sw-1(config-if)#sw mode trunk

pisa-unipi-ngaforov-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

pisa-unipi-ngaforov-sw-1(config-if)#exit
pisa-unipi-ngaforov-sw-1(config)#int f0/1
pisa-unipi-ngaforov-sw-1(config-if)#sw mode acc
pisa-unipi-ngaforov-sw-1(config-if)#sw acc vlan 401
% Access VLAN does not exist. Creating vlan 401
pisa-unipi-ngaforov-sw-1(config-if)#exit
pisa-unipi-ngaforov-sw-1(config)#vlan 401
pisa-unipi-ngaforov-sw-1(config-vlan)#name unipi-main
pisa-unipi-ngaforov-sw-1(config-vlan)#int vlan 401
pisa-unipi-ngaforov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up

pisa-unipi-ngaforov-sw-1(config-if)#no sh
pisa-unipi-ngaforov-sw-1(config-if)#exit
pisa-unipi-ngaforov-sw-1(config)#exit
pisa-unipi-ngaforov-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-unipi-ngaforov-sw-1#wr mem
Building configuration...
[OK]
pisa-unipi-ngaforov-sw-1#
```

Рис. 7: Настройка VLAN 401 и trunk-порта

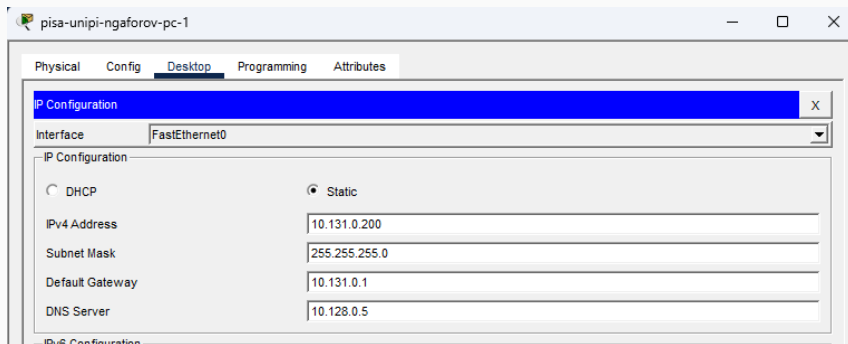


Рис. 8: Настройка IP-адреса рабочей станции

Настройка VPN

GRE-туннель на стороне «Донская»

```
msk-donskaya-ngaforov-gw-1>en
Password:
msk-donskaya-ngaforov-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-donskaya-ngaforov-gw-1(config)#
msk-donskaya-ngaforov-gw-1(config)#int tunnel 0

msk-donskaya-ngaforov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

msk-donskaya-ngaforov-gw-1(config-if)#ip addr 10.128.255.253 255.255.255.252
msk-donskaya-ngaforov-gw-1(config-if)#tunnel source f0/1.4
msk-donskaya-ngaforov-gw-1(config-if)#tunnel destination 192.0.2.20
msk-donskaya-ngaforov-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

msk-donskaya-ngaforov-gw-1(config-if)#exit
msk-donskaya-ngaforov-gw-1(config)#int loopback 0

msk-donskaya-ngaforov-gw-1(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

msk-donskaya-ngaforov-gw-1(config-if)#ip addr 10.128.254.1 255.255.255.255
msk-donskaya-ngaforov-gw-1(config-if)#exit
msk-donskaya-ngaforov-gw-1(config)#ip route 10.128.254.5 255.255.255.255 10.128.255.254
msk-donskaya-ngaforov-gw-1(config)#exit
msk-donskaya-ngaforov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-ngaforov-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-ngaforov-gw-1#
```

GRE-туннель на стороне Пизы

```
pisa-inipi-ngaforov-gw-1(config)#
pisa-inipi-ngaforov-gw-1(config)#int tunnel 0

pisa-inipi-ngaforov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

pisa-inipi-ngaforov-gw-1(config-if)#ip addr 10.128.255.254 255.255.255.252
pisa-inipi-ngaforov-gw-1(config-if)#tunnel source f0/1
pisa-inipi-ngaforov-gw-1(config-if)#tunnel destination 198.51.100.2
pisa-inipi-ngaforov-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

pisa-inipi-ngaforov-gw-1(config-if)#exit
pisa-inipi-ngaforov-gw-1(config)#int loopback 0

pisa-inipi-ngaforov-gw-1(config-if)#
%LINK-3-UPDOWN: Interface Loopback0, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

pisa-inipi-ngaforov-gw-1(config-if)#ip addr 10.128.254.5 255.255.255.255
pisa-inipi-ngaforov-gw-1(config-if)#exit
pisa-inipi-ngaforov-gw-1(config)#ip route 10.128.254.1 255.255.255.255 10.128.255.253
pisa-inipi-ngaforov-gw-1(config)#router ospf 1
pisa-inipi-ngaforov-gw-1(config-router)#router-id 10.128.254.5
pisa-inipi-ngaforov-gw-1(config-router)#net 10.0.0.0 0.255.255.255 area 0
pisa-inipi-ngaforov-gw-1(config-router)#exit
pisa-inipi-ngaforov-gw-1(config)#
00:16:01: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on Tunnel0 from LOADING to FULL, Loading Done

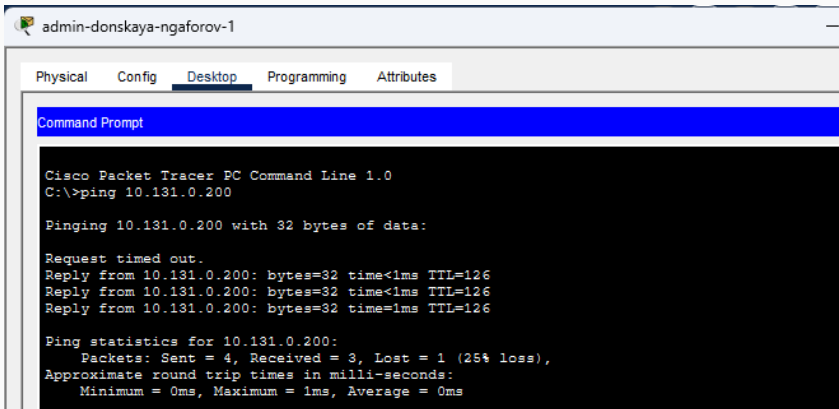
pisa-inipi-ngaforov-gw-1(config)#exit
pisa-inipi-ngaforov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-inipi-ngaforov-gw-1#wr mem
Building configuration...
[OK]
pisa-inipi-ngaforov-gw-1#
```

Проверка работы

Проверка доступности узла

- адрес назначения — 10.131.0.200
- команда **ping** получила 3 ответа из 4
- первый пакет был потерян
- последующие ответы подтвердили доступность узла



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'admin-donskaya-ngaforov-1'. The 'Desktop' tab is selected. The Command Prompt displays the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.131.0.200

Pinging 10.131.0.200 with 32 bytes of data:

Request timed out.
Reply from 10.131.0.200: bytes=32 time<1ms TTL=126
Reply from 10.131.0.200: bytes=32 time<1ms TTL=126
Reply from 10.131.0.200: bytes=32 time=1ms TTL=126

Ping statistics for 10.131.0.200:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

- 10.128.6.1 — шлюз сети «Донская»
- 10.128.255.254 — сторона GRE-туннеля в Пизе
- 10.131.0.200 — конечный узел Университета г. Пиза

Таблица маршрутизации «Донская»

- 10.131.0.0/24
- получен через OSPF
- следующий переход — 10.128.255.254
- интерфейс выхода — Tunnel0

```
msk-donskaya-ngaforov-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 29 subnets, 4 masks
C       10.128.0.0/24 is directly connected, FastEthernet0/0.3
L       10.128.0.1/32 is directly connected, FastEthernet0/0.3
C       10.128.1.0/24 is directly connected, FastEthernet0/0.2
L       10.128.1.1/32 is directly connected, FastEthernet0/0.2
C       10.128.3.0/24 is directly connected, FastEthernet0/0.101
L       10.128.3.1/32 is directly connected, FastEthernet0/0.101
C       10.128.4.0/24 is directly connected, FastEthernet0/0.102
L       10.128.4.1/32 is directly connected, FastEthernet0/0.102
C       10.128.5.0/24 is directly connected, FastEthernet0/0.103
L       10.128.5.1/32 is directly connected, FastEthernet0/0.103
C       10.128.6.0/24 is directly connected, FastEthernet0/0.104
L       10.128.6.1/32 is directly connected, FastEthernet0/0.104
C       10.128.254.1/32 is directly connected, Loopback0
S       10.128.254.5/32 [1/0] via 10.128.255.254
L       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.1/32 is directly connected, FastEthernet0/1.5
C       10.128.255.4/30 is directly connected, FastEthernet0/1.6
L       10.128.255.5/32 is directly connected, FastEthernet0/1.6
O       10.128.255.8/30 [110/2] via 10.128.255.2, 00:21:45, FastEthernet0/1.5
        [110/2] via 10.128.255.6, 00:21:45, FastEthernet0/1.6
C       10.128.255.252/30 is directly connected, Tunnel0
```

Таблица маршрутизации Университета г. Пиза

- сеть 10.131.0.0/24 подключена напрямую
- адрес шлюза 10.131.0.1 находится на FastEthernet0/0.401
- удалённые сети получены через OSPF
- маршрут по умолчанию направлен через 192.0.2.1

```
pisa-inipi-ngaforov-gw-1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is 192.0.2.1 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 20 subnets, 3 masks
O    10.128.0.0/24 [110/1001] via 10.128.255.253, 00:01:27, Tunnel0
O    10.128.1.0/24 [110/1001] via 10.128.255.253, 00:01:27, Tunnel0
O    10.128.3.0/24 [110/1001] via 10.128.255.253, 00:01:27, Tunnel0
O    10.128.4.0/24 [110/1001] via 10.128.255.253, 00:01:27, Tunnel0
O    10.128.5.0/24 [110/1001] via 10.128.255.253, 00:01:27, Tunnel0
O    10.128.6.0/24 [110/1001] via 10.128.255.253, 00:01:27, Tunnel0
S    10.128.254.1/32 [1/0] via 10.128.255.253
C    10.128.254.5/32 is directly connected, Loopback0
O    10.128.255.0/30 [110/1001] via 10.128.255.253, 00:01:27, Tunnel0
O    10.128.255.4/30 [110/1001] via 10.128.255.253, 00:01:27, Tunnel0
O    10.128.255.8/30 [110/1002] via 10.128.255.253, 00:01:27, Tunnel0
C    10.128.255.252/30 is directly connected, Tunnel0
L    10.128.255.254/32 is directly connected, Tunnel0
O    10.128.0.0/24 [110/1001] via 10.128.255.253, 00:01:27, Tunnel0
```

ИТОГ



В ходе работы:

- создана сеть Университета г. Пиза
- выполнена базовая настройка маршрутизатора и коммутатора
- настроена VLAN **401**
- назначена статическая адресация рабочей станции
- создан GRE-туннель между удалёнными сетями
- настроен обмен маршрутами через OSPF
- проверена доступность узла **10.131.0.200** из сети «Донская»